

Ch 4. Electric Fields in Matter

4.1 Polarization

$$\mathbf{p} = \alpha \mathbf{E}$$

$$\mathbf{P} \equiv \frac{d\mathbf{p}}{d\tau}$$

$$\rho_b = -\nabla \cdot \mathbf{P}$$

$$\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$$

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$$

$$U = -\mathbf{p} \cdot \mathbf{E}$$

Use: turn matter into ordinary electrostatics sources. For constant $\mathbf{P}$, $\rho_b = 0$ and only σ_b survives.

4.2 Field of a polarized object

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\oint \frac{\sigma_b}{\tau} da' + \int \frac{\rho_b}{\tau} d\tau' \right)$$

Then $\mathbf{E} = -\nabla V$.

Uniformly polarized sphere:

$$\mathbf{E}_{\text{in}} = -\frac{\mathbf{P}}{3\epsilon_0}$$

$$\mathbf{p}_{\text{sphere}} = \frac{4\pi R^3}{3} \mathbf{P}, \quad \mathbf{E}_{\text{out}} = \frac{1}{4\pi\epsilon_0 r^3} [3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}]$$

4.3 Electric displacement $\mathbf{D}$

Split charge into free + bound:

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho = \rho_f + \rho_b, \quad \rho_b = -\nabla \cdot \mathbf{P}$$

so define

$$\mathbf{D} \equiv \epsilon_0 \mathbf{E} + \mathbf{P}$$

$$\nabla \cdot \mathbf{D} = \rho_f$$

$$\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}$$

Use: with symmetry, solve for $\mathbf{D}$ from *free* charge only, then recover $\mathbf{E}$ from $\mathbf{P}$ or constitutive law.

Deceptive parallel with $\mathbf{E}$:

$$\nabla \times \mathbf{E} = 0 \quad \text{but} \quad \nabla \times \mathbf{D} = \nabla \times \mathbf{P}$$

so there is generally *no* scalar potential for $\mathbf{D}$ and no simple Coulomb-type law for $\mathbf{D}$. $\mathbf{D}$ is mainly a symmetry/Gauss-law tool.

Boundary conditions:

$$D_{\text{above}}^{\perp} - D_{\text{below}}^{\perp} = \sigma_f$$

$$\mathbf{E}_{\text{above}}^{\parallel} = \mathbf{E}_{\text{below}}^{\parallel}$$

Equivalent normal E jump:

$$\epsilon_0 (E_{\text{above}}^{\perp} - E_{\text{below}}^{\perp}) = \sigma_{\text{total}}$$

Exam move: normal component $\rightarrow$ use pillbox; tangential component $\rightarrow$ use tiny loop.

Crystal ambiguity: in ionic crystals the absolute $\mathbf{P}$ can depend on how you partition charges into dipoles; *physical* quantities are the resulting bound charges/currents and changes in polarization, not an arbitrary absolute reference for infinite crystal.

4.4 Linear dielectrics

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

$$\epsilon_r \equiv \epsilon / \epsilon_0 = 1 + \chi_e$$

Meaning: χ_e measures how easily material polarizes; ϵ_r tells how much field/capacitance changes relative to vacuum.

If homogeneous linear dielectric fills space:

$$\mathbf{E} = \mathbf{E}_{\text{vac}} / \epsilon_r$$

For anisotropic crystals:

$$P_i = \epsilon_0 \sum_j \chi_{ij} E_j$$

so $\mathbf{P}$ need not be parallel to $\mathbf{E}$.

Boundary-value problems in linear media:

$$V_1 = V_2$$

$$\epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} = \sigma_f$$

$$\epsilon_1 \frac{\partial V_1}{\partial n} - \epsilon_2 \frac{\partial V_2}{\partial n} = -\sigma_f$$

If $\sigma_f = 0$:

$$\epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}$$

$$E_1^{\parallel} = E_2^{\parallel}$$

Hence field-line bending:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_1}{\epsilon_2}$$

Inside homogeneous dielectric,

$$\rho_b = -\nabla \cdot \mathbf{P} = -\frac{\chi_e}{1 + \chi_e} \rho_f$$

so if no free charge in the bulk, then $\rho_b = 0$ in the bulk and V obeys Laplace's equation.

Capacitance/energy:

$$C = \epsilon_r C_{\text{vac}}$$

$$W = \frac{1}{2} C V^2 = \frac{1}{2} Q V = \frac{Q^2}{2C}$$

$$C_{\text{pp}} = \frac{\epsilon A}{d}$$

$$u = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon E^2 = \frac{D^2}{2\epsilon}$$

$$W_{\text{all charges}} = \frac{\epsilon_0}{2} \int E^2 d\tau$$

For linear dielectrics,

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Interpretation: this is work to assemble the *free* charge while the dielectric polarizes.

Force on dielectric slab in capacitor:

$$F = \frac{1}{2} V^2 \frac{dC}{dx} = \frac{1}{2} \frac{Q^2}{C^2} \frac{dC}{dx}$$

For slab width w and plate separation d :

$$|F| = \frac{\epsilon_0 \chi_e w}{2d} V^2$$

Direction is toward larger C (dielectric gets pulled into field).

Ch 5. Magnetostatics

5.1 Lorentz force law

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

Magnetic part exists only for moving charge and is perpendicular to both $\mathbf{v}$ and $\mathbf{B}$.

$$dW_{\text{mag}} = \mathbf{F}_{\text{mag}} \cdot d\mathbf{l} = 0$$

Meaning: magnetic fields change direction of motion, not speed/kinetic energy.

Uniform $\mathbf{B}$ trajectories:

$$R = \frac{mv_{\perp}}{|q|B}$$

$$\omega_c = \frac{|q|B}{m}$$

$$T = \frac{2\pi m}{|q|B}$$

If $v_{\parallel} \neq 0$, path is a helix with pitch $p = v_{\parallel} T$.

Currents and force on current

$$I = \frac{dq}{dt}$$

$$\mathbf{K} = \sigma \mathbf{v}$$

$$\mathbf{J} = \rho \mathbf{v}$$

For moving line charge density λ :

$$I = \lambda v$$

$$\mathbf{F} = I \int d\mathbf{l} \times \mathbf{B}$$

$$d\mathbf{F} = I d\mathbf{l} \times \mathbf{B}, \quad \mathbf{F} = \int \mathbf{K} \times \mathbf{B} da, \quad \mathbf{F} = \int \mathbf{J} \times \mathbf{B} d\tau$$

Parallel wires distance s :

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi s}$$

Same-direction currents attract; opposite directions repel. **Continuity equation:**

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$$

In magnetostatics,

$$\partial\rho/\partial t = 0, \quad \nabla \cdot \mathbf{J} = 0$$

Use: steady current cannot start/stop inside region unless charge accumulates.

5.2 Biot–Savart law

Magnetostatics requires steady currents:

$$\frac{\partial\rho}{\partial t} = 0, \quad \frac{\partial\mathbf{J}}{\partial t} = 0$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} I \int \frac{d\mathbf{l}' \times \hat{\mathbf{z}}}{\tau^2}$$

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}') \times \hat{\mathbf{z}}}{\tau^2} d\tau'$$

Also for surfaces:

$$\mathbf{B} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K} \times \hat{\mathbf{z}}}{\tau^2} da'$$

Exam warnings:

- Do *not* apply Biot–Savart to a single moving point charge in magnetostatics.
- Keep source point $\mathbf{r}'$ and field point $\mathbf{r}$ distinct.
- Direction comes from $d\mathbf{l}' \times \hat{\mathbf{z}}$ (right-hand rule).

High-frequency results from Biot–Savart/symmetry:

$$B_{\text{wire}} = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad B_{\text{loop, center}} = \frac{\mu_0 I}{2R}$$

$$B_{\text{axis of loop}}(z) = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}}$$

Finite straight segment:

$$B = \frac{\mu_0 I}{4\pi s} (\sin\theta_1 + \sin\theta_2)$$

Circular arc of angle ϕ (radians):

$$B_{\text{arc}} = \frac{\mu_0 I \phi}{4\pi R}$$

5.3 Divergence, curl, and Ampère's law

$$\nabla \cdot \mathbf{B} = 0 \quad \oint \mathbf{B} \cdot d\mathbf{a} = 0$$

Meaning: no magnetic monopoles; field lines do not begin/end.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

Derivation skeleton from Biot–Savart:

- For divergence, use $\nabla \cdot (\mathbf{J} \times \hat{\mathbf{z}}/\tau^2)$ and curls vanish.
- For curl, use $\nabla \cdot (\hat{\mathbf{z}}/\tau^2) = 4\pi\delta^3(\mathbf{z})$.
- Delta function picks out $\mathbf{J}(\mathbf{r})$.

When Ampère is useful: only when symmetry lets you pull B outside the loop.

- Infinite wire: circular loop $\Rightarrow B(2\pi s) = \mu_0 I$.
- Infinite solenoid: rectangular loop $\Rightarrow B = \mu_0 n I$ inside, ≈ 0 outside.
- Toroid: circle of radius s inside core $\Rightarrow B = \mu_0 N I / (2\pi s)$.

- Infinite current sheet with $\mathbf{K}$: $B = \mu_0 K/2$ on each side, opposite directions.

Statics comparison:

$$\nabla \cdot \mathbf{E} = \rho/\epsilon_0, \quad \nabla \times \mathbf{E} = 0 \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

5.4 Vector potential $\mathbf{A}$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

Gauge freedom:

$$\mathbf{A}' = \mathbf{A} + \nabla\lambda$$

Choose Coulomb gauge:

$$\nabla \cdot \mathbf{A} = 0$$

Then Ampère gives vector Poisson equation:

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

Integral solutions:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{\tau} d\tau'$$

$$\mathbf{A} = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}'}{\tau}$$

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{K}}{\tau} da'$$

Boundary conditions:

$$B_{\text{above}}^{\perp} = B_{\text{below}}^{\perp}$$

$$\mathbf{B}_{\text{above}}^{\parallel} - \mathbf{B}_{\text{below}}^{\parallel} = \mu_0 (\mathbf{K} \times \hat{\mathbf{n}})$$

$$\mathbf{A}_{\text{above}} = \mathbf{A}_{\text{below}},$$

$$\frac{\partial \mathbf{A}_{\text{above}}}{\partial n} - \frac{\partial \mathbf{A}_{\text{below}}}{\partial n} = -\mu_0 \mathbf{K}$$

Magnetic dipole moment of loop:

$$\mathbf{m} = I \mathbf{a}$$

Direction by right-hand rule. Far away, localized current distributions look dipolar:

$$\mathbf{A}_{\text{dip}} = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \hat{\mathbf{r}}}{r^2}$$

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}]$$

On axis: $B_{\text{axis}} = \mu_0 m / (2\pi r^3)$. In equatorial plane: $B_{\text{eq}} = \mu_0 m / (4\pi r^3)$ opposite to $\mathbf{m}$.

Ch 6. Magnetic Fields in Matter

6.1 Magnetization and magnetic dipoles

Microscopic origin: orbital motion + spin.

$$\mathbf{M} \equiv \frac{d\mathbf{m}}{d\tau}$$

Units: A/m. Material types:

- Paramagnets: $\mathbf{M}$ tends to align with field.
- Diamagnets: induced $\mathbf{M}$ opposes field.
- Ferromagnets: strong alignment + memory (history dependent).

Torque/energy/force on magnetic dipole:

$$\mathbf{N} = \mathbf{m} \times \mathbf{B} \quad U = -\mathbf{m} \cdot \mathbf{B}$$

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$$

Use: in uniform $\mathbf{B}$, net force on closed loop is zero but torque may be nonzero; translational force needs nonuniform field.

Uniform-field proof of zero net force:

$$\mathbf{F} = I \oint d\mathbf{l} \times \mathbf{B} = I \left(\oint d\mathbf{l} \right) \times \mathbf{B} = 0$$

Diamagnetism derivation (classical model):

$$\mathbf{m}_{\text{orbit}} = -\frac{1}{2} e v R \hat{\mathbf{z}}$$

Field changes orbital speed by

$$\Delta v = \frac{e R B}{2 m_e}$$

hence induced dipole moment

$$\Delta \mathbf{m} = -\frac{e^2 R^2}{4 m_e} \mathbf{B}$$

always opposite $\mathbf{B}$.

Gilbert vs Ampère model: electric-dipole analogies help intuition, but real magnets are current loops, not separated magnetic charges.

6.2 Field of a magnetized object

Build field from infinitesimal dipoles $d\mathbf{m} = \mathbf{M} d\tau'$:

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{M}(\mathbf{r}') \times \hat{\mathbf{z}}}{\tau^2} d\tau'$$

After vector identity + integration by parts:

$$\mathbf{J}_b = \nabla \times \mathbf{M}$$

$$\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$$

and the potential becomes exactly that of ordinary currents:

$$\mathbf{A} = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}_b}{r} d\tau' + \frac{\mu_0}{4\pi} \oint \frac{\mathbf{K}_b}{r} da'$$

Use: replace magnetized matter by equivalent bound volume/surface current and then solve with Biot-Savart/Ampère.

Shortcut: if $\mathbf{M}$ is uniform, then $\mathbf{J}_b = 0$ and only $\mathbf{K}_b = \mathbf{M} \times \hat{\mathbf{n}}$ remains.

$$|\mathbf{K}_b| = M \text{ for side surface of uniformly magnetized cylinder}$$

Uniformly magnetized sphere:

$$\mathbf{m}_{\text{sphere}} = \frac{4\pi R^3}{3} \mathbf{M}$$

$$\mathbf{H}_{\text{in}} = -\frac{\mathbf{M}}{3}$$

$$\mathbf{B}_{\text{in}} = \frac{2}{3}\mu_0 \mathbf{M}$$

6.3 Auxiliary field $\mathbf{H}$

Total current splits into free + bound:

$$\mathbf{J} = \mathbf{J}_f + \mathbf{J}_b, \quad \mathbf{J}_b = \nabla \times \mathbf{M}$$

Define

$$\mathbf{H} \equiv \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$$

Then

$$\nabla \times \mathbf{H} = \mathbf{J}_f$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{f,\text{enc}}$$

Use: when symmetry is set by *free current*, solve for $\mathbf{H}$ first, then recover $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$.

Deceptive parallel with $\mathbf{D}$:

$$\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$$

so $\mathbf{H}$ is generally not divergenceless; there is no general Biot-Savart law for $\mathbf{H}$.

Boundary conditions:

$$B_{\text{above}}^{\perp} - B_{\text{below}}^{\perp} = 0$$

$$\mathbf{H}_{\text{above}}^{\parallel} - \mathbf{H}_{\text{below}}^{\parallel} = \mathbf{K}_f \times \hat{\mathbf{n}}$$

Equivalent normal jump for H :

$$H_{\text{above}}^{\perp} - H_{\text{below}}^{\perp} = -(M_{\text{above}}^{\perp} - M_{\text{below}}^{\perp})$$

If no free surface current, tangential $\mathbf{H}$ is continuous.

6.4 Linear and nonlinear media

For linear magnetic media:

$$\mathbf{M} = \chi_m \mathbf{H}$$

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu \mathbf{H}$$

$$\mu = \mu_0(1 + \chi_m)$$

$$\mu_r = \mu/\mu_0 = 1 + \chi_m$$

Signs: $\chi_m > 0$ for paramagnets, $\chi_m < 0$ for diamagnets. If no free surface current at interface of linear media:

$$B_1^{\perp} = B_2^{\perp} \quad \frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

In homogeneous linear media:

$$\mathbf{J}_b = \nabla \times \mathbf{M} = \chi_m \nabla \times \mathbf{H} = \chi_m \mathbf{J}_f$$

So if $\mathbf{J}_f = 0$ in bulk, then $\mathbf{J}_b = 0$ in bulk too; bound current lives on surface.

Ferromagnets (nonlinear):

- Domains align; response is not $\mathbf{M} = \chi_m \mathbf{H}$ except maybe locally.
- Saturation: beyond some H , M stops increasing much.
- Hysteresis: B depends on history.
- Residual magnetization: magnetization left after $H \rightarrow 0$.
- Coercive field: reverse H needed to demagnetize.
- Curie point: above T_C , ferromagnet becomes paramagnetic.

Fast Problem-Solving Map

- **Given $\mathbf{P}$:** compute ρ_b, σ_b , then use electrostatics.

- **Given dielectric + free charge symmetry:** use $\oint \mathbf{D} \cdot d\mathbf{a} = Q_{f,\text{enc}}$.
- **Given steady current geometry:** try Biot-Savart first; switch to Ampère if symmetry is high.
- **Given magnetized matter:** replace by $\mathbf{J}_b, \mathbf{K}_b$.
- **Given magnetic material with free-current symmetry:** solve $\mathbf{H}$, then $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$ or $\mathbf{B} = \mu \mathbf{H}$ if linear.
- **Boundary problem:** normal component usually comes from divergence law; tangential component from loop/Ampère law.
- **Derivation points:** write the defining field equation first, then the theorem used (Gauss/Stokes/integration by parts), then the symmetry argument.